Label-Free, Dependency-Aware Channel Selection for Coupled Multi-Channel Sensor Data

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Introduction

Complex sensors deliver dozens to hundreds of channels that overlap heavily and arrive in natural groups: inertial units, excitation wavelengths, scalp regions, process subsystems.

For cost, power, bandwidth and interpretability we often want to keep a handful of the actual channels rather than a learned mixture of them.

The hard part is deciding **which** channels to keep when no class labels are available.

Problem Statement

Unsupervised filters such as variance ranking or PCA loadings score each channel by its own **marginal** statistics: fast, but blind to how channels depend on one another, and biased toward the loudest signal, which is not always the informative one.

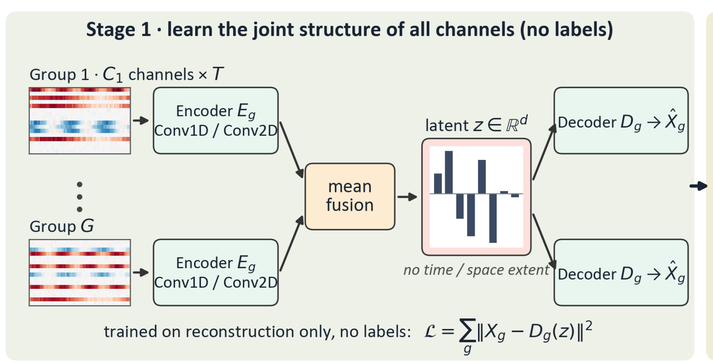
Supervised dependency-aware selectors (mRMR, JMI) model conditional relevance but need labels.

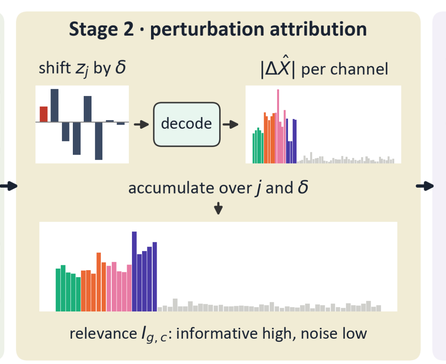
Few methods are at once **label-free, dependency-aware and discrete**, returning real channels rather than a projection.

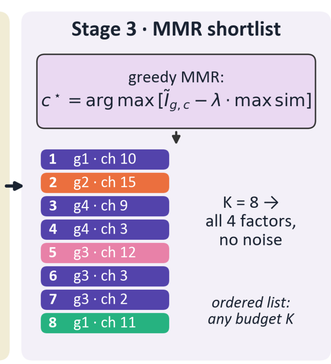
Novelty

A label-free, dependency-aware selector that returns real channels. A group-structured autoencoder with a **compressive bottleneck** learns the joint structure of all channels; **perturbing each latent factor** and measuring the per-channel reconstruction change gives a label-free relevance; a **relevance–redundancy rule** turns it into an ordered shortlist. Only the encoder changes between modalities. Its channels match a supervised selector and beat PCA and variance where the discriminative signal is not the loudest.

Method: compressive group autoencoder → perturbation attribution → relevance–redundancy selection







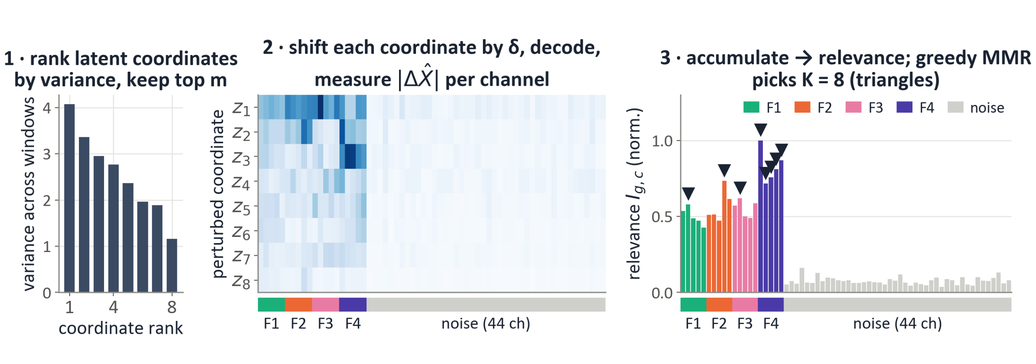


Figure 1: The three-stage pipeline with real intermediate outputs from the controlled benchmark: per-group encoders and mean fusion give a compressive code z; shifting each latent coordinate and decoding attributes a reconstruction change to every channel; MMR orders the channels.

Figure 2: Attribution on the benchmark: latent coordinates ranked by variance (1), per-channel reconstruction change of each coordinate (2), accumulated relevance with the eight MMR picks (3).

1 · Compressive group autoencoder

Each group is encoded independently (Conv1D for signals, Conv2D for images); the features are averaged into a latent code z with no time or space extent and decoded per group. Training uses reconstruction error only.

$$z := \frac{1}{G}\sum_{g=1}^{G} E_g(X_g), \qquad \hat{X}_g := D_g(z), \qquad \mathcal{L} := \sum_{g=1}^{G} \|X_g - D_g(z)\|_2^2$$

2 · Perturbation attribution

Latent coordinates are ranked by variance and the top m kept; each is shifted by magnitudes δ, and the accumulated per-channel change of the decoded output, weighted by its variance rank w_j, is the label-free relevance.

$$I_{g,c} := \sum_{j=1}^{m}\sum_{\delta\in\mathcal{D}} w_j \frac{1}{NT}\sum_{n=1}^{N}\sum_{t=1}^{T} \left|\left[D_g(z+\delta e_j) - D_g(z)\right]_{n,t,c}\right|$$

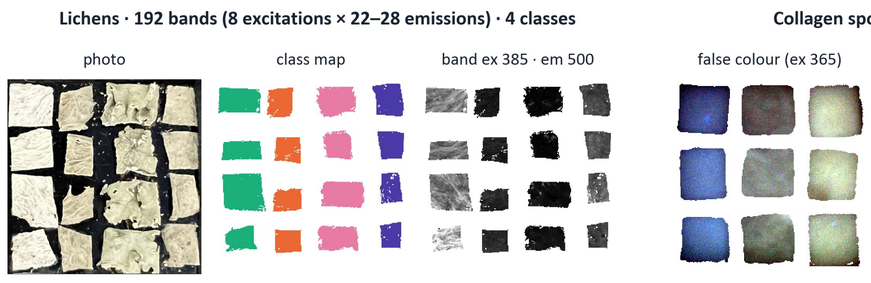
3 · Relevance–redundancy selection

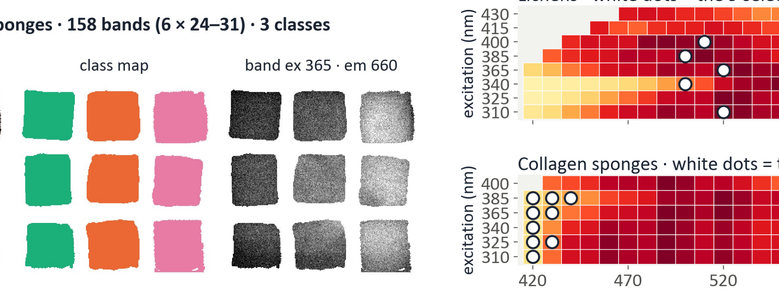
Relevances are normalised within each group; channels are picked greedily by maximal marginal relevance (relevance minus λ × cosine similarity to channels already picked). The ordered output truncates to any budget K.

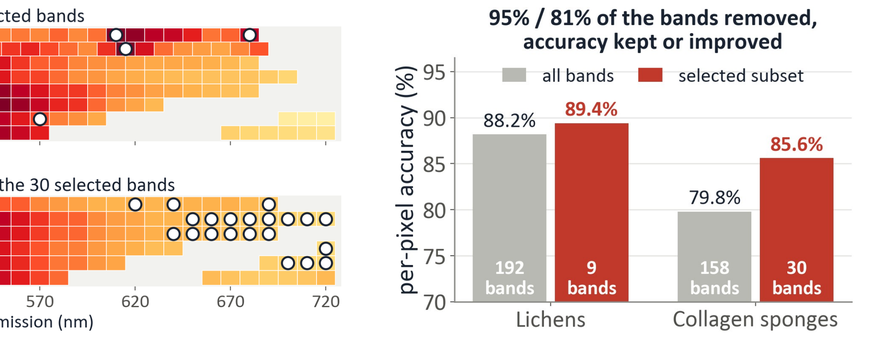
$$c^\star = \arg\max_{(g,c)\notin S} \left[\tilde{I}_{g,c} - \lambda \max_{s\in S} \text{sim}\big((g,c),s\big)\right], \qquad \tilde{I}_{g,c} := \frac{I_{g,c}}{\max_{c'} I_{g,c'}}$$

Protocol: *selection is always unsupervised. Labels are used only by a downstream classifier that scores the selected K channels (accuracy or macro-F1) against variance ranking, PCA loadings, a supervised mutual-information (MI) selector and random-K.*

Controlled benchmark: four latent factors hidden among 64 channels





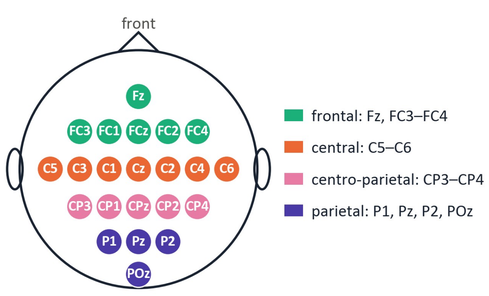


(a) 64 grouped time-series channels: four independent latent factors, each in a cluster of five correlated channels of decreasing amplitude, plus 44 pure-noise channels. The class (16 = 2^4 factor states) is decodable only by covering all four factors.

(b) kNN accuracy vs K. Eight label-free channels beat the full noisy set (0.94 vs 0.87) and match the supervised selector; PCA also recovers the factors, which here are the loudest channels; variance piles into one cluster (0.24).

(c) The channels each selector keeps at K = 8: the label-free method covers all four factors with no noise channel. (The bottleneck occasionally under-encodes a factor: 0.70 ± 0.29 mean accuracy over random draws of the benchmark.)

Motor-imagery EEG (BCI Competition IV-2a): the informative rhythm is not the loudest signal

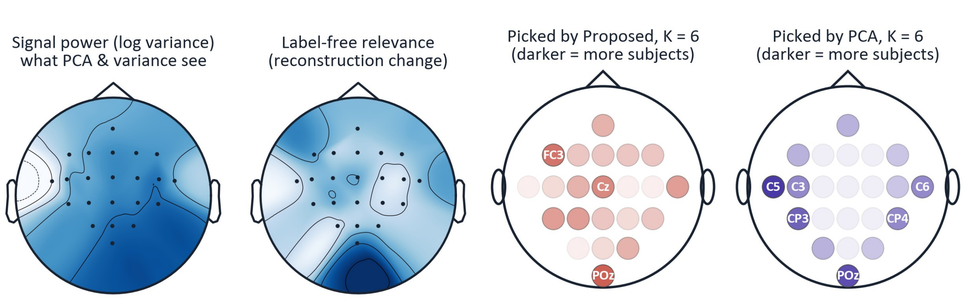


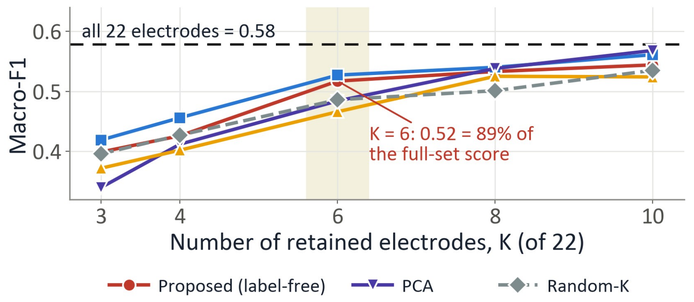
22 electrodes in 4 scalp-region groups

4 classes: left hand, right hand, feet, tongue

9 subjects · 250 Hz · band-pass 8 –30 Hz

Protocol: selection + CSP/LDA fit on session 1, test on session 2 (within-subject)





(a) Montage and groups: the grouping is acquisition metadata, not labels.

(b) Why marginal methods fail here: signal power peaks over lateral and occipital electrodes (muscle, alpha, ocular activity), so variance and PCA pick loud but uninformative electrodes (C5, C6, POz); the reconstruction-based relevance concentrates on the sensorimotor strip (FC3, Cz, C1, CP1, CP3).

(c) Macro-F1 vs K, mean over nine subjects. At six of 22 electrodes the label-free method reaches 0.517 (89% of the full 0.578), above PCA, variance and random and close to supervised MI.

Biomedical multi-excitation fluorescence imaging: up to 95% of the bands removed without loss

(a) Multi-excitation fluorescence cubes: for each excitation wavelength a full emission spectrum is recorded per pixel, so every excitation–emission pair is one image and one channel: 192 bands for lichens (4 classes), 158 for collagen sponges (3 classes). The band images shown are among the selected ones.

(b) Where the selected bands sit on the excitation × emission grid (background: mean emission per excitation, row-normalised). The picks spread across excitations instead of clustering on the brightest bands.

(c) 9 of 192 lichen bands: 89.4% vs 88.2% with all bands (a 95% reduction); 30 of 158 collagen bands: 85.6% vs 79.8%. Pruning also denoises.

Results across regimes: where label-free selection pays, and where nothing beats a random subset

Selector	Synthetic accuracy, K = 8 / 64	EEG macro-F1, K = 6 / 22
All channels (full set)	0.87	0.58
Supervised MI	0.99	0.53
Proposed (label-free)	**0.94**	**0.52**
PCA	0.95	0.48
Variance	0.24	0.47
Random-K	0.25	0.49

Table 1: Downstream classifier score on the selected channels; no labels are used for selection. The label-free method stays within 0.05 (synthetic) and 0.01 (EEG) of the supervised selector.

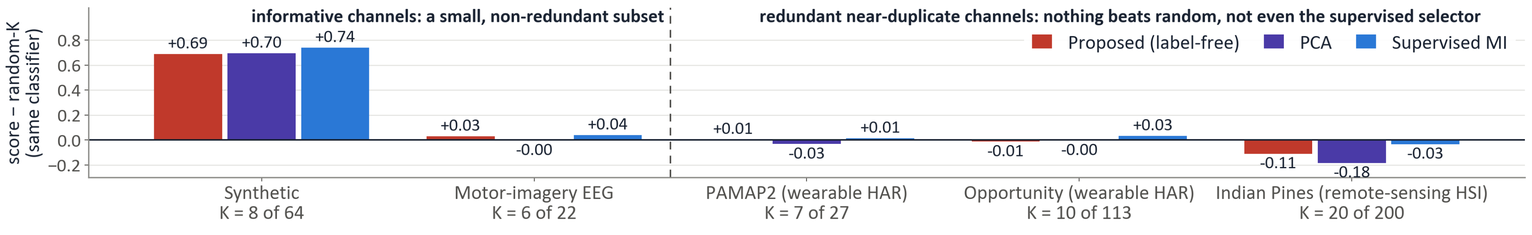


Figure 3: Gain over a random subset of the same size. Selection pays where the informative channels are a small, non-redundant fraction of the input; where channels are near-duplicates (wearable HAR, remote sensing) not even the supervised selector beats random, and a label-using check flags this in advance.

Conclusion and Next Steps

- A compressive group autoencoder, perturbation attribution and a relevance–redundancy rule select real channels without labels; only the encoder changes between modalities.
- The selected channels match a supervised selector and beat PCA and variance where the discriminative signal is not the loudest; up to 95% of imaging bands removed without loss.
- A simple label-using check tells in advance when any selection is pointless.

Next: tighten the bottleneck, make the attribution less sensitive to channel variance, scale to larger channel arrays.

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paper code